

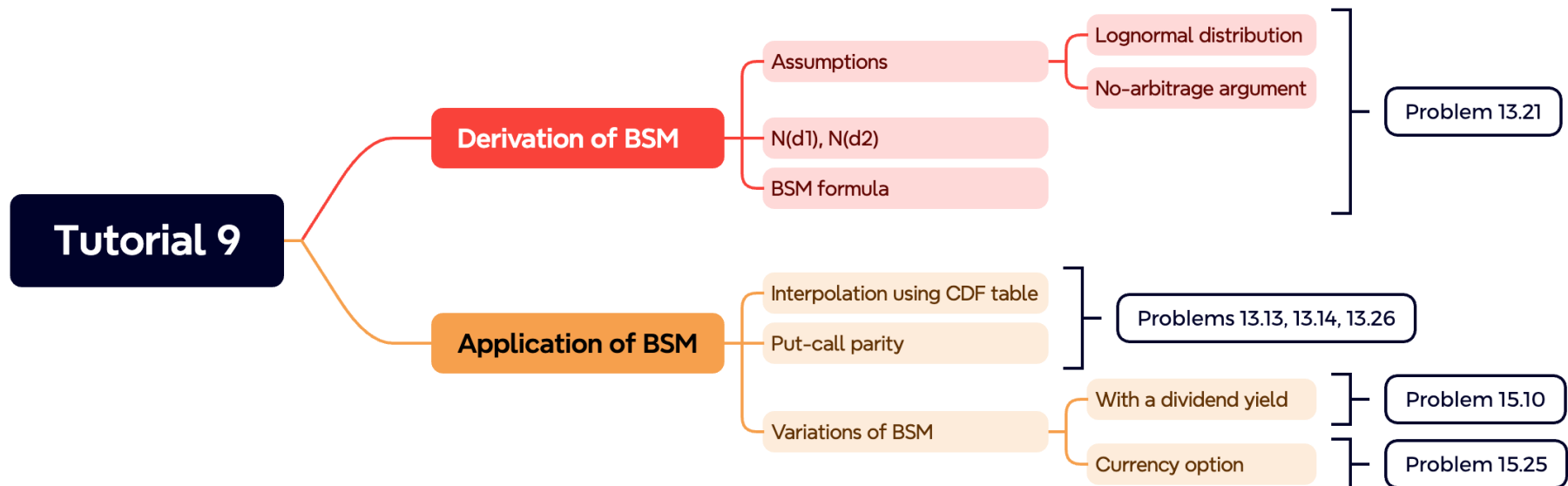
FINA2204 Tutorial 9: The Black-Scholes-Merton Model

Zirui Song

UWA Business School

Oct 2026

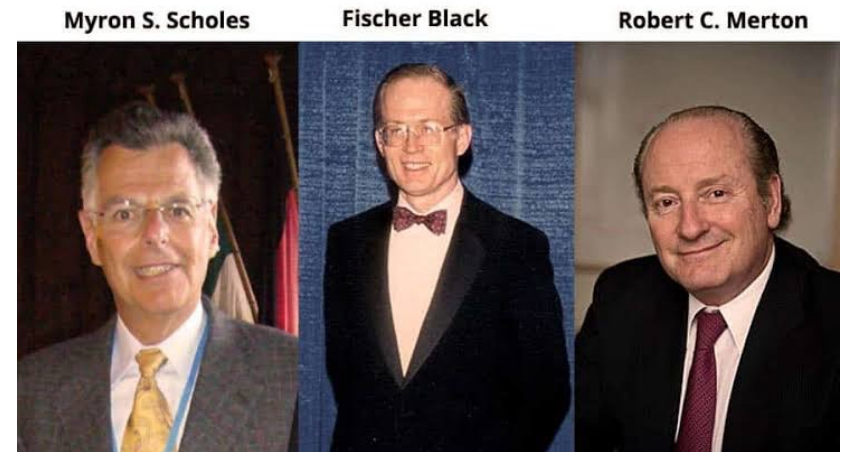
Agenda



Presented with xmind

Derivation of BSM: the big picture

- In 1973, Black, Scholes and Merton derived a formula to price options.
- The paper had been rejected by other journals before the *Journal of Political Economy* (JPE) accepted it.
- By 1975, traders were using calculators with the formula built in.



Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1997

Summary

Laureates

Robert C. Merton

Myron Scholes

Press release

Advanced information

Award ceremony speech

Share this



Press release

English

Swedish



14 October 1997

[The Royal Swedish Academy of Sciences](#) has decided to award **the Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel, 1997**, to

Professor **Robert C. Merton**, Harvard University, Cambridge, USA and
Professor **Myron S. Scholes**, Stanford University, Stanford, USA

for a new method to determine the value of derivatives.

Robert C. Merton and Myron S. Scholes have, in collaboration with the late Fischer Black, developed a pioneering formula for the valuation of stock options. Their methodology has paved the way for economic valuations in many areas. It has also generated new types of financial instruments and facilitated more efficient risk management in society.

Derivation of BSM: the big picture

Same insight as the binomial tree

- The binomial model prices options using replication and no-arbitrage (μ does not matter $\rightarrow$ price as if the stock earns r)

Change: shrink the time step

- As the time intervals become smaller, the stock-price process becomes continuous (lognormal).

$$S_T = S_0 e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}, \quad Z \sim \phi(0,1)$$

- BSM = the binomial model with infinitely many tiny steps.

Derivation of BSM: the big picture

- An option price is the present value of its expected payoff under **the risk-neutral probability measure, Q** :

$$c = e^{-rT} E^Q [\max(S_T - K, 0)]$$

- For a call option, the payoff is positive only when $S_T > K$. So when we work out this expectation, it splits into two pieces:

$$c = \underbrace{S_0 N(d_1)}_{\text{PV of stock you receive}} - \underbrace{K e^{-rT} N(d_2)}_{\text{PV of strike you pay}}$$

where $N(d_2)$ = the risk-neutral probability of exercise; $N(d_1)$ counts the same exercise states, but each one is weighted by S_T .

Derivation of BSM: the big picture

- The put has the same structure in reverse.

$$p = e^{-rT} E^Q [\max(K - S_T, 0)]$$

$$p = \underbrace{e^{-rT} K N(-d_2)}_{\text{PV of strike you receive}} - \underbrace{S_0 N(-d_1)}_{\text{PV of stock you give up}}$$

- BSM satisfies put-call parity**, as any no-arbitrage model must!

$$c - p = [S_0 N(d_1) - K e^{-rT} N(d_2)] - [e^{-rT} K N(-d_2) - S_0 N(-d_1)]$$

$$= S_0 - K e^{-rT}$$

Derivation of BSM: assumptions

The assumptions made by Black, Scholes, and Merton when they derived their option pricing formula were as follows:

1. Stock price behavior corresponds to the lognormal model (developed earlier in this chapter) with μ and σ constant.
2. There are no transaction costs or taxes. All securities are perfectly divisible.
3. There are no dividends on the stock during the life of the option.
4. There are no riskless arbitrage opportunities.
5. Security trading is continuous.
6. Investors can borrow or lend at the same risk-free rate of interest.
7. The short-term risk-free rate of interest, r , is constant.

Some of these assumptions have been relaxed by other researchers. For example, variations on the Black–Scholes–Merton formula can be used when r and σ are functions of time and, as we shall see later in this chapter, the formula can be adjusted to take dividends into account.

Derivation of BSM

- The binomial tree represents stock-price uncertainty using **discrete** up-and-down movements.
- As $\Delta t \rightarrow 0$, discrete random steps become **continuous** random movements.
- The random part is **Brownian motion** W . Over a short interval Δt :
$$\Delta W = Z\sqrt{\Delta t}, \quad Z \sim \phi(0, 1)$$
$$\rightarrow \Delta W \sim \phi(0, \Delta t)$$
- The change has mean 0, variance Δt , and standard deviation $\sqrt{\Delta t}$.
- **But why a normal distribution?**

Derivation of BSM: Galton Board

“A wonderful form of cosmic order.”

- A device invented by statistician Francis Galton (1873).
- Each ball = **a random walk** through a binomial tree: one left/right step per row.
- Many random steps add up to **a normal distribution**.
- More rows, tinier steps $\rightarrow$ Brownian motion, with $\Delta W \sim \phi(0, \Delta t)$.



Derivation of BSM

A stock option pricing model must make some assumptions about how stock prices evolve over time. If a stock price is \$100 today, what is the probability distribution for the price in one day or in one week or in one year?

The Black–Scholes–Merton model considers a non-dividend-paying stock and assumes that the return on the stock in a very short period of time, Δt , is normally distributed. The returns in two different nonoverlapping periods are assumed to be independent. If we define

μ : Expected return on the stock

σ : Volatility of the stock price

then the mean of the return in time Δt is $\mu \Delta t$ and the standard deviation of the return in time Δt is $\sigma\sqrt{\Delta t}$. The Black–Scholes–Merton assumption is therefore that

$$\frac{\Delta S}{S} \sim \phi(\mu \Delta t, \sigma^2 \Delta t) \quad (13.1)$$

where ΔS is the change in the stock price S in time Δt , and $\phi(m, v)$ denotes a normal distribution with mean m and variance v . Note that it is the variance of the return, not its standard deviation, that is proportional to Δt .

The Lognormal Distribution

When relatively long time periods are considered, it is necessary to be more precise about the future stock price distribution. The assumption in equation (13.1) implies that the stock price at any future time has a *lognormal* distribution. The general shape of a lognormal distribution is shown in Figure 13.1. It can be contrasted with the more familiar normal distribution in Figure 13.2. Whereas a variable with a normal distribution can take any positive or negative value, a lognormally distributed variable is restricted to being positive. A normal distribution is symmetrical; a lognormal distribution is skewed with the mean, median, and mode all different.

skewed to the right

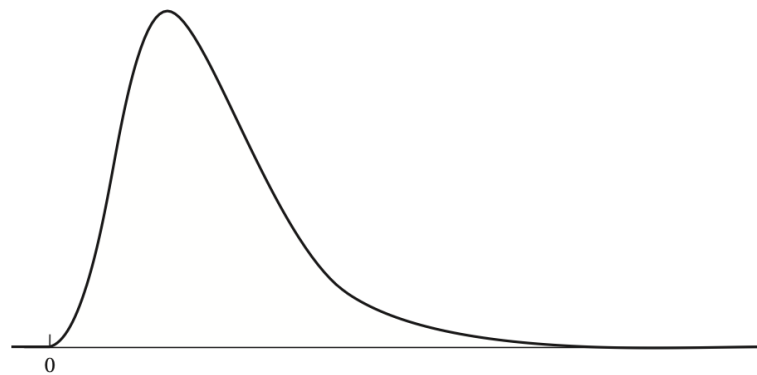


Figure 13.1 A lognormal distribution

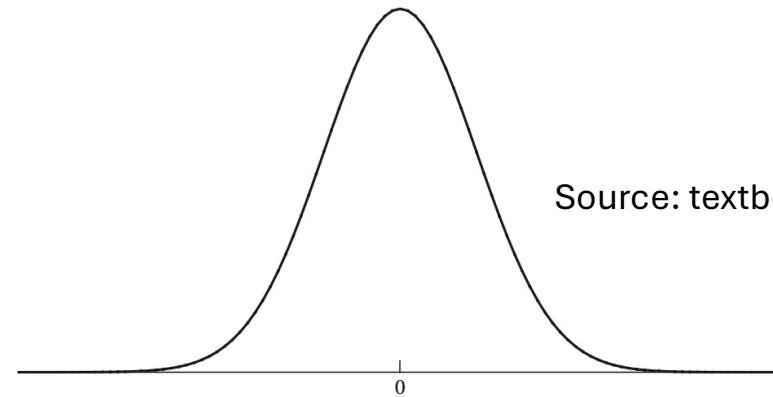
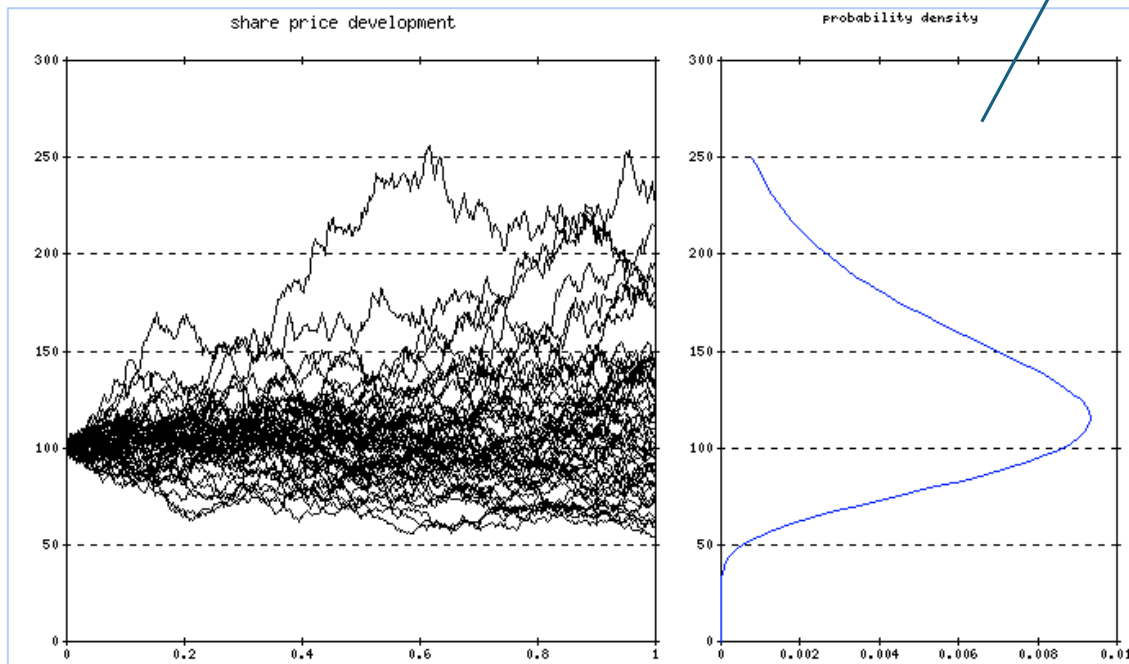


Figure 13.2 A normal distribution

Source: textbook, p.313

Simulations of Geometric Brownian Motion

Lognormal distribution!



basic share data			
drift	0.2	volatility	0.35
initial value	100	time (years)	1
indications			
☐	show moving average	in days	38
☐	show confidence interval	probability	0.75
☐	show expectation		
☐	show median		
estimation using historical data			
☐	show drift and vol	over days	30
extended configuration			
no. of realizations	50		
image size x	400	image size y	400
☐	init random generator	with number	825

9/10/2026

FINA2204 Tutorial 9

Source: <https://kluge.in-chemnitz.de/tools/sharesim/>₂

Derivation of BSM

When relatively long time periods are considered, it is necessary to be more precise about the future stock price distribution. The assumption in equation (13.1) implies that the stock price at any future time has a *lognormal* distribution. The general shape of a lognormal distribution is shown in Figure 13.1. It can be contrasted with the more familiar normal distribution in Figure 13.2. Whereas a variable with a normal distribution can take any positive or negative value, a lognormally distributed variable is restricted to being positive. A normal distribution is symmetrical; a lognormal distribution is skewed with the mean, median, and mode all different.

A variable with a lognormal distribution has the property that its natural logarithm is normally distributed. The Black–Scholes–Merton assumption for stock prices therefore implies that $\ln S_T$ is normal, where S_T is the stock price at a future time T . The mean and standard deviation of $\ln S_T$ can be shown to be

S_T is lognormal $\rightarrow \ln S_T$ is normal

$$\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T \quad \text{and} \quad \sigma\sqrt{T}$$

where S_0 is the current stock price. We can write this result as

$$\ln S_T \sim \phi \left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right] \quad (13.2)$$

Derivation of BSM

- Therefore,

$$\ln S_T = \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z, \quad Z \sim \phi(0, 1)$$

- Taking the exponential of both sides:

$$e^{\ln S_T} = e^{\ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z}$$

Since

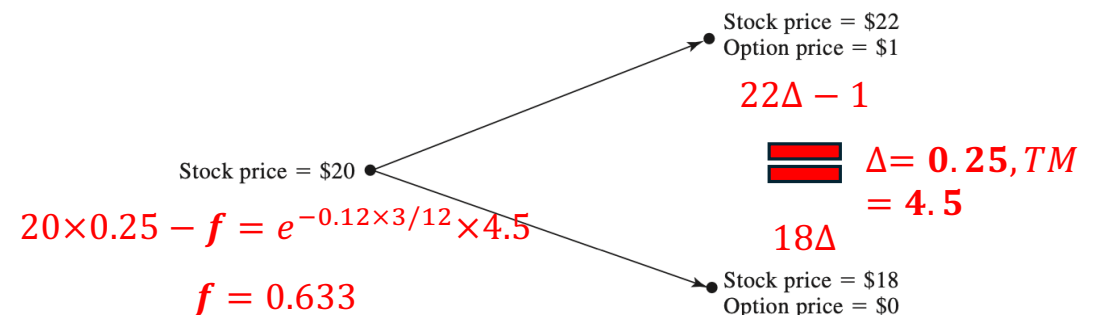
$$e^{\ln S_T} = S_T$$

We obtain:

$$S_T = e^{\ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z} = e^{\ln S_0} \cdot e^{\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z} = S_0 e^{\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z}$$

Derivation of BSM

- In the real world, the stock price grows on average at μ .
- But μ depends on investors' expectations and risk appetites.
- The binomial model showed us **we don't need it**.
- Using replication and no arbitrage, we can instead price the option in a risk-neutral world.



Derivation of BSM

- In the real world:

$$S_T = S_0 e^{\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}$$

- For option pricing, replace:

Real world: $\mu \rightarrow$ Risk – neutral world: r

- Therefore, under the risk-neutral measure Q :

$$S_T = S_0 e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}$$

- The drift changes from μ to r ; the volatility remain σ .

Derivation of BSM

- At expiration, a European call pays:
 $\max(S_T - K, 0)$
- In the risk-neutral world, calculate the expected payoff.
- Then discount it back to today at the risk-free rate (r).

Therefore:

$$c = e^{-rT} E^Q[\max(S_T - K, 0)]$$

- The Q reminds us that the expectation is being calculated under **the risk-neutral probability measure**.

Derivation of BSM

- The call has a positive payoff **only when:**

$$S_T > K$$

so:

$$\begin{aligned} S_T &= S_0 e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z} > K \\ \rightarrow e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z} &> \frac{K}{S_0} \end{aligned}$$

Take logs:

$$\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z > \ln\left(\frac{K}{S_0}\right)$$

Derivation of BSM

Therefore:

$$Z > \frac{\ln\left(\frac{K}{S_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

Define

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

Hence:

$$S_T > K \leftrightarrow \mathbf{Z} > -\mathbf{d}_2$$

What is the probability
of that happening?

Recap: Derivation of BSM

- The function $N(x)$ is the **cumulative distribution function (CDF)** for a variable with a standard normal distribution.
- Graphically, it is the shaded area to the left of x .

so,

$$N(x) = Q(Z \leq x)$$

$N(0.5)=0.6915?$

$N(-0.5)=0.3085?$

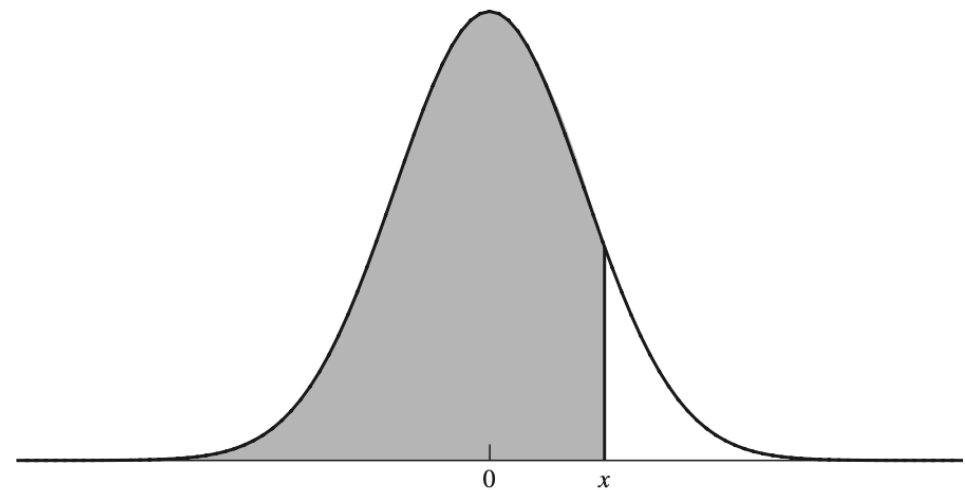


Figure 13.4 Shaded area represents $N(x)$

Source: textbook, p.322

Recap: Derivation of BSM

Properties of $N(x)$

1. It is a probability.

$$0 \leq N(x) \leq 1$$

• So

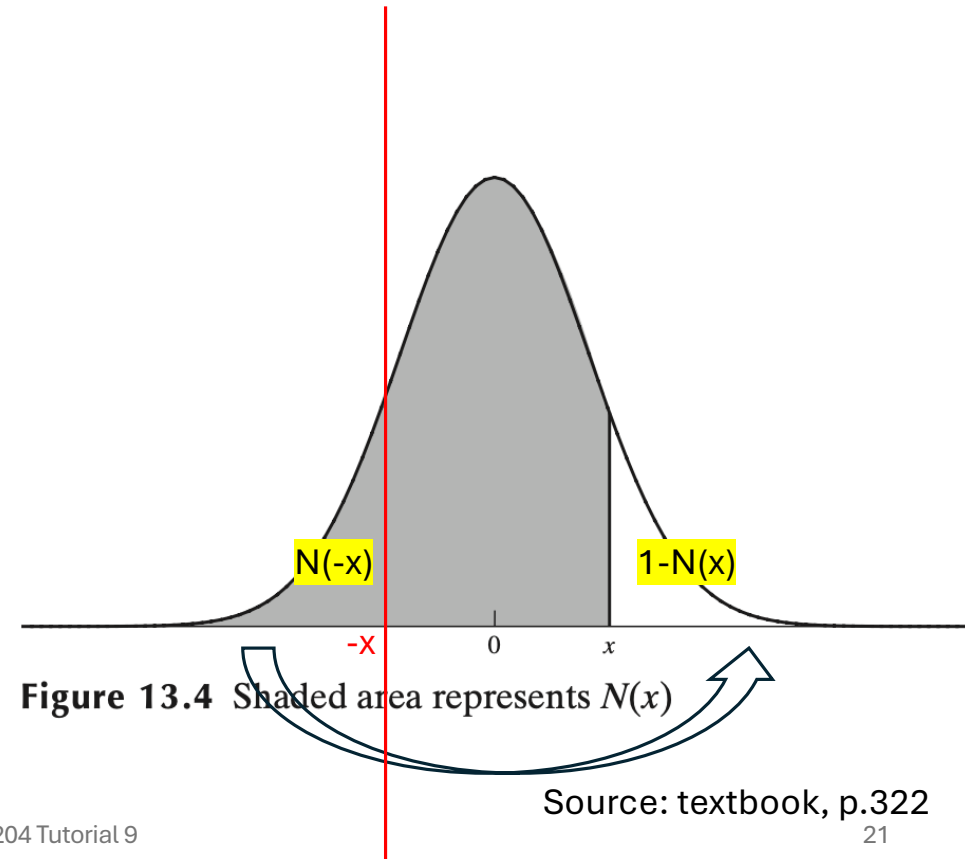
$$1 - N(x) = Q(Z > x)$$

• area above = 1 - area below

2. Mirror image (symmetric)

• area left of $-x$ = right of x

$$N(x) = 1 - N(-x)$$



Recap: Derivation of BSM

- The curve itself is the **probability density function (PDF)** of the standard normal:

$$N'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Adding up all strips from minus infinity to x , you get the CDF.

$$N(x) = \int_{-\infty}^x N'(x) dx$$

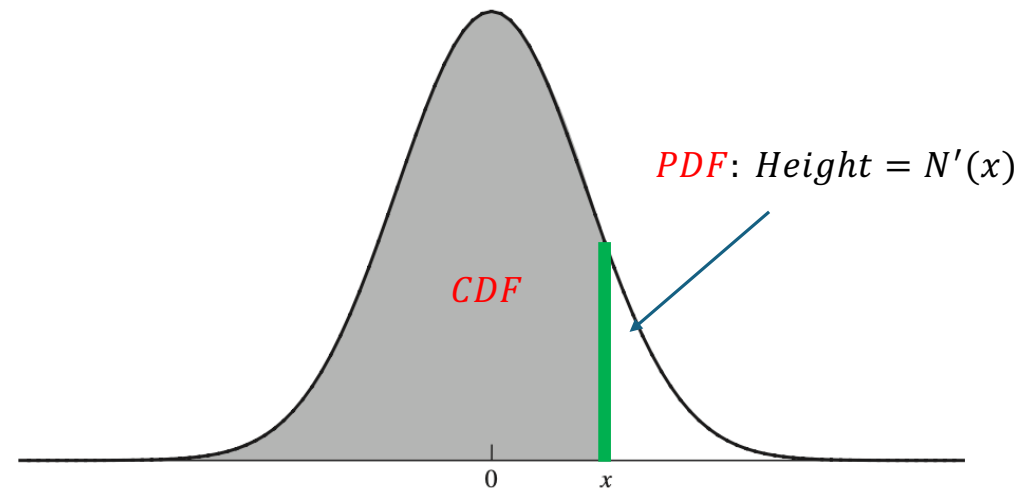


Figure 13.4 Shaded area represents $N(x)$

Table for $N(x)$ When $x \geq 0$

This table shows values of $N(x)$ for $x \geq 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(0.6278) &= N(0.62) + 0.78[N(0.63) - N(0.62)] \\ &= 0.7324 + 0.78 \times (0.7357 - 0.7324) \\ &= 0.7350 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

$N(0.53)=?$

first decimal place

second decimal place

Derivation of BSM

- Exercise condition: $S_T > K \leftrightarrow Z > -d_2$
- Same event, so same probability:
$$Q(S_T > K) = Q(Z > -d_2)$$
- Since $\mathbf{Z} \sim \boldsymbol{\phi}(\mathbf{0}, \mathbf{1})$, using the cumulative distribution function $N(\cdot)$:
$$Q(Z > -d_2) = 1 - N(-d_2)$$
- Because the standard normal distribution is symmetric,
$$1 - N(-d_2) = N(d_2)$$

Therefore,

$$Q(S_T > K) = N(d_2)$$

- $N(d_2)$ is the risk-neutral probability that the call will be exercised!

Derivation of BSM

- The call is exercised ($S_T > K$) with probability $N(d_2)$.
- Otherwise, if $S_T < K$, it pays **nothing**.
- Conditional on exercise, the payoff is $(S_T - K)$.
- Therefore, the **expected payoff** can be written as:

$$E^Q[\max(S_T - K, 0)] = \underbrace{E^Q[S_T - K | S_T > K]}_{\text{average payoff when exercised}} \underbrace{N(d_2)}_{\text{probability of exercise}}$$

Problem 13.21

- *Show that the probability that a European call option will be exercised in a risk-neutral world is, with the notation introduced in this chapter, $N(d_2)$.*
- *What is an expression for the value of a derivative that pays off \$100 if the price of a stock at time T is greater than K ?*

Answer 13.21

Part 1: probability of exercise

Stock price under Q:

$$S_T = S_0 e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}$$

The call is exercised if $S_T > K$.

$$S_0 e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z} > K$$

Divide by S_0 and take logs:

$$\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\mathbf{Z} > \ln \frac{K}{S_0}$$

Answer 13.21

Solve for Z :

$$Z > \frac{\ln\left(\frac{K}{S_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} = -\frac{\ln\left(\frac{K}{S_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} = -d_2$$

Hence

$$S_T > K \leftrightarrow Z > -d_2$$

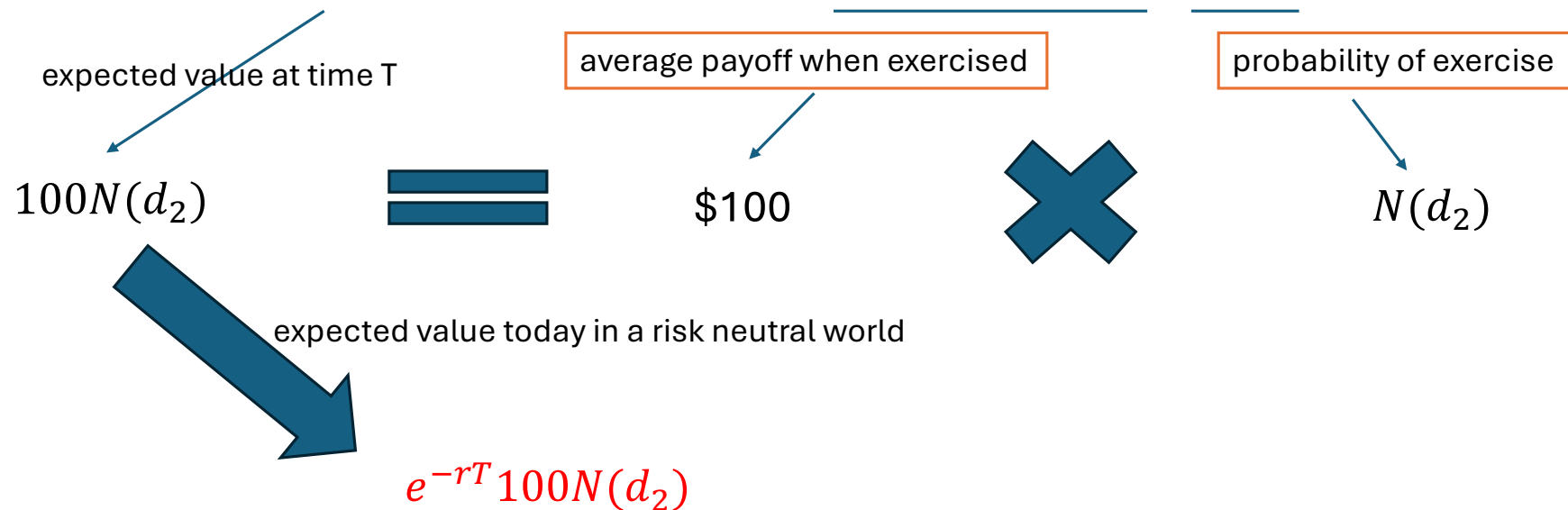
Using $N(x) = Q(Z \leq x)$ and the mirror property $N(x) = 1 - N(-x)$:

$$Q(S_T > K) = N(d_2)$$

Answer 13.21

Part 2: value of the \$100 derivative

$$E^Q[\max(S_T - K, 0)] = E^Q[S_T - K | S_T > K] N(d_2)$$



Derivation of BSM

- Split the term inside the expectation:

$$\begin{aligned} E^Q[\max(S_T - K, 0)] &= [E^Q(S_T | S_T > K) - K]Q(S_T > K) \\ &= E^Q(S_T | S_T > K)Q(S_T > K) - KQ(S_T > K) \end{aligned}$$

- $E^Q[\max(S_T - K, 0)] = \underbrace{E^Q(S_T | S_T > K)Q(S_T > K)}_{\text{stock you receive}} - \underbrace{KQ(S_T > K)}_{\text{strike you pay}}$

?

- We already know:

$$Q(S_T > K) = N(d_2)$$

- Therefore, the strike-price part is:

$$KQ(S_T > K) = KN(d_2)$$

Derivation of BSM

- Goal: compute **the stock-price part**

$$E^Q(S_T | S_T > K) Q(S_T > K)$$

- Since $S_T > K \leftrightarrow Z > -d_2$; $Q(S_T > K) = Q(Z > -d_2)$

So we can rewrite everything in terms of Z .

$$E^Q(S_T | S_T > K) Q(S_T > K) = E^Q(S_T | Z > -d_2) Q(Z > -d_2)$$

- Z is continuous, so a thin strip at z has probability $N'(z)dz$.

Derivation of BSM

- Adding up all the strips to the right of $-d_2$ gives an **integral**:

$$(S_T | Z > -d_2) Q(Z > -d_2) \\ = \int_{-d_2}^{\infty} S_T(z) N'(z) dz = S_0 e^{(r - \frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} e^{\sigma\sqrt{T}z} N'(z) dz$$

- Since $N'(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, $\sigma\sqrt{T}z - \frac{1}{2}z^2 = -\frac{1}{2}(z - \sigma\sqrt{T})^2 + \frac{1}{2}\sigma^2T$

$$\rightarrow S_0 e^{(r - \frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} e^{\sigma\sqrt{T}z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$= S_0 e^{(r - \frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma\sqrt{T}z - \frac{1}{2}z^2} dz = S_0 e^{rT} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2} dz$$

Derivation of BSM

- Compare the integrand with the standard normal PDF:

$$N'(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \text{ vs. } N'(z - \sigma\sqrt{T}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2}$$

- The only difference: z is replaced by $z - \sigma\sqrt{T}$, so the integrand is

$$N'(z - \sigma\sqrt{T}).$$

- A normal density that's been shifted to the right!

So

$$\rightarrow \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2} dz = \int_{-d_2}^{\infty} N'(z - \sigma\sqrt{T}) dz$$

Derivation of BSM

- Let $y = z - \sigma\sqrt{T}$, so $dy = dz$.
- Change the limits:

Lower limit: $z = -d_2 \rightarrow y = -d_2 - \sigma\sqrt{T}$,

Upper limit: $z = \infty \rightarrow y = \infty$

Therefore,

$$\rightarrow \int_{-d_2}^{\infty} N'(\textcolor{red}{z} - \textcolor{red}{\sigma}\sqrt{\textcolor{red}{T}})dz = \int_{\textcolor{blue}{-d_2} - \textcolor{blue}{\sigma}\sqrt{\textcolor{blue}{T}}}^{\infty} N'(\textcolor{red}{y})dy$$

Derivation of BSM

- Now define

$$d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

so the lower limit is $-d_1$.

- The integral is now an area under the standard normal curve.

$$\rightarrow \int_{-d_1}^{\infty} N'(y)dy = 1 - N(-d_1) = N(d_1)$$

Derivation of BSM

- Putting it together, the stock-price part equals:

$$E^Q(S_T | S_T > K) Q(S_T > K)$$

$$= E^Q(S_T | Z > -d_2) Q(Z > -d_2) = S_0 e^{rT} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma\sqrt{T})^2} dz$$

$$= S_0 e^{rT} \int_{-d_2}^{\infty} N'(z - \sigma\sqrt{T}) dz = S_0 e^{rT} \int_{-d_1}^{\infty} N'(y) dy = S_0 e^{rT} N(d_1)$$

Derivation of BSM

- Now we have both pieces.

- Thus, the expected payoff can be written as:

$$\begin{aligned} E^Q(S_T | S_T > K) Q(S_T > K) - K Q(S_T > K) \\ = S_0 e^{rT} N(d_1) - K N(d_2) \end{aligned}$$

- Discount back to today at risk-free rate:

$$c = e^{-rT} [S_0 e^{rT} N(d_1) - K N(d_2)]$$

Therefore:

$$c = \underbrace{S_0 N(d_1)}_{\text{PV of stock you receive}} - \underbrace{K e^{-rT} N(d_2)}_{\text{PV of strike you pay}}$$

Derivation of BSM

The term $N(d_2)$ in equation (13.5) has a fairly simple interpretation. It is the probability that a call option will be exercised in a risk-neutral world. The $N(d_1)$ term is not quite so easy to interpret. The expression $S_0 N(d_1) e^{rT}$ is the expected stock price at time T in a risk-neutral world when stock prices less than the strike price are counted as zero. The strike price is only paid if the stock price is greater than K and as just mentioned this has a probability of $N(d_2)$. The expected payoff in a risk-neutral world is therefore

$$S_0 N(d_1) e^{rT} - K N(d_2)$$

Present-valuing this from time T to time zero gives the Black–Scholes–Merton equation for a European call option:

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

For another way of looking at the Black–Scholes–Merton equation for the value of a European call option, note that it can be written as

$$c = e^{-rT} N(d_2) [S_0 e^{rT} N(d_1) / N(d_2) - K]$$

The terms here have the following interpretation:

e^{-rT} : Present value factor

$N(d_2)$: Probability of exercise

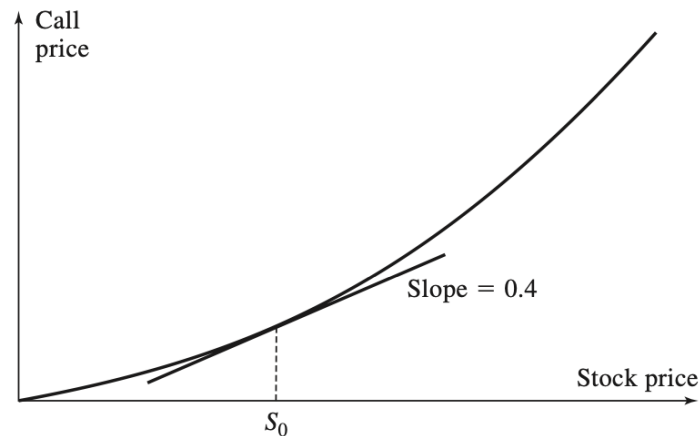
$S_0 e^{rT} N(d_1) / N(d_2)$: Expected stock price in a risk-neutral world if option is exercised

K : Strike price paid if option is exercised.

$$E^Q(S_T | S_T > K) = E^Q(S_T) N(d_1) / N(d_1) \\ \rightarrow N(d_1) = N(d_2) \times \frac{E^Q(S_T | S_T > K)}{E^Q(S_T)}$$

In words: It counts the same exercise states as $N(d_2)$, but weighted by the stock price you receive.

Derivation of BSM



$$N(d_1) = \text{delta} = \frac{\partial C}{\partial S}!$$

Figure 13.3 Relationship between call price and stock price. Current stock price is S_0

The reason a riskless portfolio can be set up is that the stock price and the option price are both affected by the same underlying source of uncertainty: stock price movements. In any short period of time, the price of a call option is perfectly positively correlated with the price of the underlying stock; the price of a put option is perfectly negatively correlated with the price of the underlying stock. In both cases, when an appropriate portfolio of the stock and the option is set up, the gain or loss from the stock position always offsets the gain or loss from the option position so that the overall value of the portfolio at the end of the short period of time is known with certainty.

Source: textbook, p.321

Derivation of BSM

- At expiration, a **European put** pays:

$$\max(K - S_T, 0)$$

- The put has a positive payoff only when:

$$S_T < K \leftrightarrow Z < -d_2$$

- Split the expected payoff, exactly as for the call:

$$E^Q[\max(K - S_T, 0)] = KQ(S_T < K) - E^Q(S_T | S_T < K)Q(S_T < K)$$

- **Strike part:**

$$KQ(S_T < K) = KQ(Z < -d_2) = KN(-d_2)$$

Derivation of BSM

- **Stock part** (Let $y = z - \sigma\sqrt{T}$):

$$E^Q(S_T | S_T < K) Q(S_T < K) = S_0 e^{rT} \int_{-\infty}^{-d_1} N'(y) dy = S_0 e^{rT} N(-d_1)$$

Therefore:

$$p = e^{-rT} [KN(-d_2) - S_0 e^{rT} N(-d_1)] = e^{-rT} KN(-d_2) - S_0 N(-d_1)$$

Derivation of BSM

- BSM is consistent with **put-call parity!**

$$\begin{aligned} c - p &= [S_0 N(d_1) - K e^{-rT} N(d_2)] - [e^{-rT} K N(-d_2) - S_0 N(-d_1)] \\ &= S_0 - K e^{-rT} \end{aligned}$$

The components of portfolio A are worth c and Ke^{-rT} today, and the components of portfolio C are worth p and S_0 today. Hence,

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

This relationship is known as *put-call parity*. It shows that the value of a European

Derivation of BSM

The Black–Scholes–Merton formulas for the prices of European calls and puts on non-dividend-paying stocks are⁶

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad (13.5)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \quad (13.6)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

S_0
 K
 $r \rightarrow c, p?$
 σ
 T

Source: textbook, p.322

Table for $N(x)$ When $x \geq 0$

This table shows values of $N(x)$ for $x \geq 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(0.6278) &= N(0.62) + 0.78[N(0.63) - N(0.62)] \\ &= 0.7324 + 0.78 \times (0.7357 - 0.7324) \\ &= \underline{0.7350} \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

Q1:

$N(0.12)=?$

A:

$N(0.12)=0.5478$

Q2:

$N(-0.12)=?$

A:

$N(x) = 1 - N(-x)$ (the mirror property)

$1 - N(0.12) = 1 - 0.5478 = \mathbf{0.4522}$

Q3:

$N(0.6278)=?$

Linear interpolation approach (problem involving **four** decimal places)

$$N(x) = N(x_1) + \frac{x - x_1}{x_2 - x_1} [N(x_2) - N(x_1)]$$

note: x is the intermediate value to be estimated. x_1 and x_2 ($|x_2| \geq |x_1|$) are the adjacent known values from the table

Source: textbook, p.601

Problem 13.13

- *What is the price of a **European call option** on a non-dividend-paying stock when the stock price is \$52, the strike price is \$50, the risk-free interest rate is 12% per annum, the volatility is 30% per annum and the time to maturity is three months?*

Answer 13.13

Inputs

- $S_0 = \$52$
- $K = \$50$
- $\sigma = 30\%$
- $r = 12\%$
- $T = 3/12$

A European call option (c?)

Answer 13.13

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad (13.5)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \quad (13.6)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$d_1 = \frac{\ln\left(\frac{52}{50}\right) + \left(0.12 + \frac{0.3^2}{2}\right) \times 0.25}{0.30\sqrt{0.25}} = 0.5365$$
$$d_2 = d_1 - 0.30\sqrt{0.25} = 0.3865$$

Answer 13.13

$$N(x) = N(x_1) + \frac{x - x_1}{x_2 - x_1} [N(x_2) - N(x_1)]$$

- According to the linear interpolation approach

$$N(0.5365) = N(0.53) + \frac{0.5365 - 0.53}{0.54 - 0.53} \times (N(0.54) - N(0.53))$$

$$= 0.7019 + 0.65 \times (0.7054 - 0.7019)$$

$$N(0.3865) = N(0.38) + 0.65 \times (N(0.39) - N(0.38))$$

$$= 0.6480 + 0.65 \times (0.6517 - 0.6480)$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389

Answer 13.13

$$N(d_1) = N(0.5365) = 0.7042$$

$$N(d_2) = N(0.3865) = 0.6504$$

- Following equation (13.5), the price of the European call is

$$c = 52 \times 0.7042 - 50e^{-0.03} \times 0.6504 = 5.06$$

or \$5.06

Full process: list inputs and identify what type of option you have, calculate d_1 and d_2 , find the CDF values, then substitute into the BSM formula.

Problem 13.14

- *What is the price of a **European put option** on a non-dividend-paying stock when the stock price is \$69, the strike price is \$70, the risk-free interest rate is 5% per annum, the volatility is 35% per annum and the time to maturity is six months?*

Answer 13.14

Inputs

- $S_0 = \$69$
- $K = \$70$
- $\sigma = 35\%$
- $r = 5\%$
- $T = 6/12$

A European put option (p?)

Answer 13.14

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad (13.5)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \quad (13.6)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$d_1 = \frac{\ln\left(\frac{69}{70}\right) + \left(0.05 + \frac{0.35^2}{2}\right) \times 0.5}{0.35\sqrt{0.5}} = 0.1666$$

$$d_2 = d_1 - 0.35\sqrt{0.5} = -0.0809$$

Answer 13.14

$$N(x) = N(x_1) + \frac{x - x_1}{x_2 - x_1} [N(x_2) - N(x_1)]$$

- According to the linear interpolation approach

$$N(-d_1) = 1 - N(d_1) = 1 - N(0.1666)$$

$$\begin{aligned} N(0.1666) &= N(0.16) + 0.66 \times (N(0.17) - N(0.16)) \\ &= 0.5636 + 0.66 \times (0.5675 - 0.5636) = 0.5662 \end{aligned}$$

$$\text{So } N(-d_1) = 1 - 0.5662 = 0.4338$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389

Answer 13.14

$$N(-d_2) = N(0.0809) = N(0.08) + 0.09 \times (N(0.09) - N(0.08))$$

$$= 0.5319 + 0.09 \times (0.5359 - 0.5319) = 0.5323$$

- The price of the European put is

$$p = 70e^{-0.025} \times 0.5323 - 69 \times 0.4338 = 6.40$$

or \$6.40.

<i>x</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389

Problem 13.26

- *Consider an option on a non-dividend-paying stock when the stock price is \$30, the exercise price is \$29, the risk-free interest rate is 5% per annum, the volatility is 25% per annum and the time to maturity is four months.*
- *a) What is the price of the option if it is a European call?*
- *b) What is the price of the option if it is an American call?*
- *c) What is the price of the option if it is a European put?*
- *d) Verify that put–call parity holds.*

Answer 13.26

Inputs

- $S_0 = \$30$
- $K = \$29$
- $r = 5\%$
- $\sigma = 25\%$
- $T = 4/12$

An option on a non-dividend-paying stock ($c = C$)

Answer 13.26

a)

where

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad (13.5)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \quad (13.6)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

A European call

$$d_1 = \frac{\ln\left(\frac{30}{29}\right) + \left(0.05 + \frac{0.25^2}{2}\right) \times \frac{4}{12}}{0.25\sqrt{0.3333}} = 0.4225$$
$$d_2 = d_1 - 0.25\sqrt{0.3333} = 0.2782$$

Ans **Table for $N(x)$ When $x \geq 0$**

This table shows values of $N(x)$ for $x \geq 0$. The table should be used with interpolation. For example,

a)

$$\begin{aligned} N(0.6278) &= N(0.62) + 0.78[N(0.63) - N(0.62)] \\ &= 0.7324 + 0.78 \times (0.7357 - 0.7324) \\ &= 0.7350 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879

A European call

$$d_1 = \frac{\ln\left(\frac{30}{29}\right) + \left(0.05 + \frac{0.25^2}{2}\right) \times \frac{4}{12}}{0.25\sqrt{0.3333}} = 0.4225$$

$$d_2 = d_1 - 0.25\sqrt{0.3333} = 0.2782$$

Answer 13.26

$$\begin{aligned}N(d_1) &= N(0.4225) = 0.6637 \\N(-d_1) &= 1 - N(0.4225) = 0.3363 \\N(d_2) &= N(0.2782) = 0.6096 \\N(-d_2) &= 1 - N(0.2782) = 0.3904\end{aligned}$$

The European call price is

$$c = 30 \times 0.6637 - 29e^{-0.05 \times 4/12} \times 0.6096 = 2.52$$

Or \$2.52

b) The American call price **is the same as** the European call price ($c = C$). It is \$2.52.

Answer 13.26

c) The European put price is

$$p = 29e^{-0.05 \times 4/12} \times 0.3904 - 30 \times 0.3363 = 1.05$$

or \$1.05

d) Put-call parity states that

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

- In this case, $c = 2.52$, $S_0 = 30$, $K = 29$, $p = 1.05$, and $e^{-rT} = 0.9835$ and it is easy to verify that the relationship is satisfied.

Variations: with a dividend yield

By replacing S_0 by $S_0 e^{-qT}$ in the Black–Scholes–Merton formulas (13.5) and (13.6), we obtain the price c of a European call and the price p of a European put on a stock paying a dividend yield at rate q as

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \quad (15.4)$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1) \quad (15.5)$$

Since

$$\ln \frac{S_0 e^{-qT}}{K} = \ln \frac{S_0}{K} - qT$$

it follows that d_1 and d_2 are given by

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

These results were first derived by Merton.³ As discussed in Chapter 13, the word *dividend* should, for the purposes of option valuation, be defined as the reduction in the stock price on the ex-dividend date arising from any dividends declared. If the dividend yield rate is known but not constant during the life of the option, equations (15.4) and (15.5) are still true, with q equal to the average dividend yield per year (measured with continuous compounding) during the option's life.

S_0
 K
 r
 σ
 T
 $+$
 q

Variations of the put-call parity

1. European (non-dividend-paying stock)

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

2. European (dividend-paying stock, a known **dollar dividend**)

$$c + \underline{D} + Ke^{-rT} = p + S_0 \quad (10.10)$$

3. European (dividend-paying stock, a known **dividend yield**)

$$??? \quad \longrightarrow \quad c + Ke^{-rT} = p + \underline{S_0 e^{-qT}} \quad (15.3)$$

Replace S_0 by $S_0 e^{-qT}$

Notice: no σ in any version of put-call parity! Because parity comes purely from no-arbitrage, not from any model of how the stock moves.

Problem 15.10

- Consider a stock index currently standing at 250. *The dividend yield on the index is 4% per annum* and the risk-free rate is 6% per annum. A three-month *European call option* on the index with a strike price of 245 is currently worth \$10. What is the value of a three-month **put option on the index** with a strike price of 245?

Answer 15.10

Inputs

- $S_0 = \$250$

- $K = \$245$

- $r = 0.06$

- $T = \frac{3}{12}$

- $q = 0.04$

- $c = 10$

σ is unknown $\rightarrow$ can't use BSM directly

A put option on the index (p?)

Answer 15.10

- Using put-call parity

$$c + Ke^{-rT} = p + S_0e^{-qT} \quad (15.3)$$

Substituting

$$p = 10 + 245e^{-0.06 \times 0.25} - 250e^{-0.04 \times 0.25} = 3.84$$

The put price is \$3.84

Variations: valuation of currency option

To value currency options, we define S_0 as the spot exchange rate. To be precise, S_0 is the value of one unit of the foreign currency in U.S. dollars. As explained in Section 5.10, a foreign currency is analogous to a stock paying a known dividend yield. The owner of foreign currency receives a yield equal to the risk-free interest rate, r_f , in the foreign currency. Inequalities (15.1) and (15.2), with q replaced by r_f , provide bounds for the European call price, c , and the European put price, p :

$$c \geq \max(S_0 e^{-r_f T} - K e^{-r T}, 0)$$

$$p \geq \max(K e^{-r T} - S_0 e^{-r_f T}, 0)$$

Equation (15.3), with q replaced by r_f , provides the put–call parity result for European currency options:

$$c + K e^{-r T} = p + S_0 e^{-r_f T}$$

Finally, equations (15.4) and (15.5) provide the pricing formulas for European currency options when q is replaced by r_f :

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-r T} N(d_2) \quad (15.8)$$

$$p = K e^{-r T} N(-d_2) - S_0 e^{-r_f T} N(-d_1) \quad (15.9)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r - r_f + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - r_f - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

S_0
 K
 r
 σ
 T
 $+$
 r_f

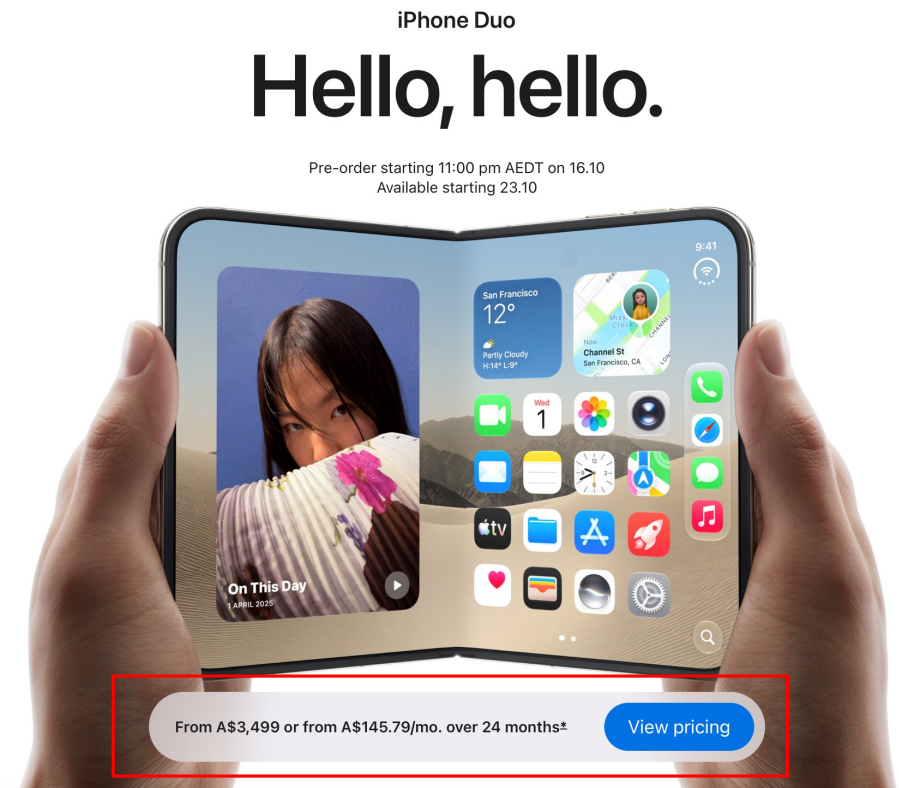
Foreign or domestic? A price tag

Price tag: 1 iPhone Duo = A\$3,499

- **iPhone** = the thing you buy; A\$ = the money you pay with

Exchange rate, same logic!

- 1 **USD** = 1.2 AUD or $\text{USD}/\text{AUD} = 1.2$
- USD = the thing you buy → foreign
- AUD = the money you pay with → domestic
- $S_0 = 1.2$ (AUD per USD)



Foreign or domestic? A price tag

- Exam-style prompt: *“Suppose that the spot price of the US dollar is AUD 1.2 ...The risk-free rates of interest in Australia and the United States are 3% and 5% per annum, respectively.”*

$$1 \text{ USD} = 1.2 \text{ AUD}$$

Foreign good

domestic money

	USD	AUD
Role	What you buy	What you pay with
Type	Foreign currency	Domestic currency
Rate	$r_f = 5\%$	$r = 3\%$

Variations of the put-call parity

1. European (non-dividend-paying stock)

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

2. European (dividend-paying stock, a known dollar dividend)

$$c + \underline{D} + Ke^{-rT} = p + S_0 \quad (10.10)$$

3. European (dividend-paying stock, a known dividend yield)

$$c + Ke^{-rT} = p + S_0 e^{-qT} \quad (15.3)$$

4. European (Foreign Currency Option)

???



$$c + Ke^{-rT} = p + \underline{S_0 e^{-r_f T}}$$

Replace S_0 by $S_0 e^{-r_f T}$

Problem 15.25

- *Suppose that the spot price of the Canadian dollar is USD 0.95 and that the Canadian dollar/US dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. (1) Calculate the value of a **European call option to buy one Canadian dollar** for USD 0.95 in nine months. (2) Use put–call parity to calculate the price of a **European put option** to sell one Canadian dollar for USD 0.95 in nine months. (3) What is the price of a call option to buy USD 0.95 with one Canadian dollar in nine months?*

Answer 15.25

Inputs

- $K = 0.95$
- $\sigma = 0.08$
- $T = \frac{9}{12}$
- $S_0 = 0.95$
- $r_f = 4\%$
- $r = 5\%$



Price tag?

$$\underline{1 \text{ CAD}} = \underline{0.95 \text{ USD}}$$

Foreign good

domestic money

CAD is the foreign currency, because it is the thing being priced.

USD is the domestic currency, because it is what we pay.

Answer 15.25

(1)

	$c = S_0 e^{-r_f T} N(d_1) - K e^{-r T} N(d_2)$	(15.8)
	$p = K e^{-r T} N(-d_2) - S_0 e^{-r_f T} N(-d_1)$	(15.9)
where	$d_1 = \frac{\ln(S_0/K) + (r - r_f + \sigma^2/2)T}{\sigma\sqrt{T}}$ $d_2 = \frac{\ln(S_0/K) + (r - r_f - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$	

$$d_1 = \frac{\ln\left(\frac{0.95}{0.95}\right) + \left(0.05 - 0.04 + \frac{0.08^2}{2}\right) \times \frac{9}{12}}{0.08\sqrt{9/12}} = 0.1429$$

$$d_2 = d_1 - 0.08\sqrt{0.75} = 0.0736$$

Answer 15.25

$$N(d_1) = N(0.1429) = 0.5568$$

$$N(d_2) = N(0.0736) = 0.5293$$

The value of the call is given by

$$c = 0.95 \times e^{-0.04 \times 0.75} \times 0.5568 - 0.95 \times e^{-0.05 \times 0.75} \times 0.5293 = 0.029$$

or 2.9 cents.

(2) From

Equation (15.3), with q replaced by r_f , provides the put-call parity result for European currency options:

$$c + Ke^{-rT} = p + S_0 e^{-r_f T}$$

$$\text{So that } p = 0.029 + 0.95e^{-0.05 \times \frac{9}{12}} - 0.95e^{-0.04 \times \frac{9}{12}} = 0.0221$$

Answer 15.25

(3) The option to buy USD 0.95 with CAD 1.00 is **the same as** an option to sell one Canadian dollar for USD 0.95 (*put on CAD*). This means that it is a put option on the Canadian dollar and its price is USD 0.0221.

The right to buy USD 0.95 with CAD 1 = the right to sell CAD 1 for USD 0.95